

TANNER L. CABANISS

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EDUCATION

- Ph.D.** Mechanical Engineering, University of Oklahoma Expected May 2027
- Dissertation (in-progress): *Development of Personalized Shape Memory Polymer (SMP)-Based Scaffolds for the Endovascular Embolization of Unruptured Cerebral Aneurysms*
- B.S.** Mechanical Engineering, Minor in Mathematics, University of Oklahoma May 2023

RESEARCH EXPERIENCE

- Graduate Research Assistant**, Biomechanics and Biomaterials Design Lab Norman, OK
Supervisor: Dr. Chung-Hao Lee May 2023 – present
- Refined SMP scaffolds via radiopaque filler integration and centrifugation to exceed 98% porosity.
 - Developed in-house testing equipment for application specific in-vitro evaluations of SMP scaffolds.
 - Mentored three undergraduate projects including camera- and PIV-based integrated flow loop systems.
 - Established a digital twin pipeline for in-silico hemodynamics evaluations of implanted SMP scaffolds.
- Undergraduate Research Assistant** Aug. 2021 – May 2023
- Devised a sacrificial templating technique for the replication of patient-specific aneurysm geometries.
 - Performed thermomechanical characterizations via FTIR, DSC, and DMA for SMP scaffolds.

SELECTED PUBLICATIONS (OUT OF 6 TOTAL PUBLICATIONS)

Cabaniss, T.L., Colby, G., et al. (2026). *In-Vitro Evaluations of Shape Memory Polymer Scaffolds with Tunable Architecture for the Endovascular Embolization of Unruptured Intracranial Aneurysms*. J. Biomech. Eng. doi.org/10.1115/1.4071060

Vega, S.R., Mesa, J.C., Blanquer, C., Barinaga, C., **Cabaniss, T.L.**, et al. (2025). *Towards the Intravascular Delivery of 3D-printed Leached SMP for Endovascular Treatment of Intracranial Aneurysm*. Ann. Biomed. Eng. doi.org/10.1007/s10439-025-03816-w

Cabaniss, T.L., Bodlak, R., et al. (2024). *CFD Investigations of a Shape-Memory Polymer Foam-Based Endovascular Embolization Device*. Biomech. Model. Mechanobiol. doi.org/10.1007/s10237-024-01910-x

SELECTED PRESENTATIONS (OUT OF 15 TOTAL PRESENTATIONS)

“Development of High-Porosity Radiopaque Shape Memory Polymer Scaffolds for Personalized Cerebral Aneurysm Therapy.” 10th World Congress of Biomechanics. 2026.

“Shape Memory Polymer Scaffolds with Tunable Architecture for the Endovascular Treatment of Cerebral Aneurysms.” Univ. of Okla. 3 Minute Thesis Competition. Advanced to Midwest Association of Graduate School Competition. 2025.

“Advancing Shape Memory Polymer Metamaterials for the Effective Treatment of Complex, Irregular Intracranial Aneurysms.” American Society of Mechanical Engineers SBC PhD Student Paper Competition. 2025.

“Performative Characterization of Shape Memory Polymer Scaffolds for Endovascular Cerebral Aneurysm Therapeutics.” American Society of Mechanical Engineers SBC. 2025.

HONORS AND AWARDS

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| 2025 | 3 Minute Thesis Competition Winner, University of Oklahoma |
| 2025 | ASME SBC PhD Student Paper Competition, 3 rd Place in Category |
| 2023 | SB ³ C Bachelor of Science Student Paper Competition Finalist |
| 2022 | Provost's URECA Summer Fellowship |

TECHNICAL SKILLS

ANSYS Fluent, SolidWorks, Blender, Python, C++, MATLAB, R, Arduino/MCUs, Manual milling, GD&T, Design for Manufacturing, FTIR, DSC, DMA, SEM, Instron